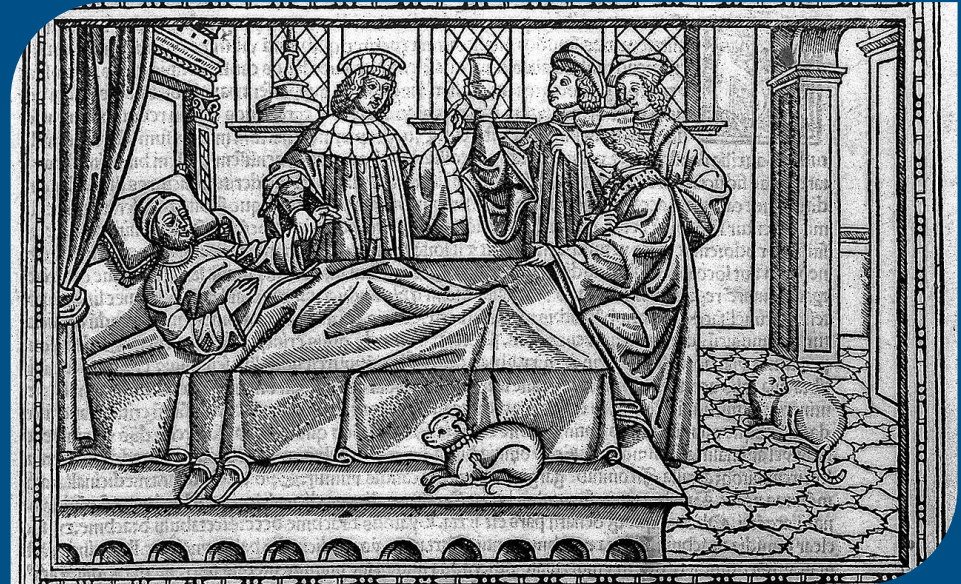


Project Aethelgard: Decentralized Clinical Intelligence via Federated RAG

GOAI-2026 Challenge

Aliaksei Kaliutau

© 2026 Monodromy



01 The Diagnostic Odyssey

The average rare disease diagnosis takes 4.8 years and 7.3 physicians. The answers exist, but they are trapped in silos.

ID	Age (years) at interview	Condition	Gender	Location (Australia)	Location classification	Time to diagnosis (years)	Year of diagnosis
001	30–39	Myositis	Female	QLD	Major cities	3–4	2015–19
002	70–79	Myositis	Male	ACT	Major cities	10 plus	2015–19
003	60–69	Sarcoidosis	Female	NSW	Inner regional	10 plus	2015–19
004	60–69	PID	Female	NSW	Major cities	10 plus	2005–09
005	40–49	PID	Female	NSW	Major cities	3–4	2015–19
006	50–59	Sarcoidosis	Female	NSW	Major cities	5–7	2010–14
007	60–69	Myositis	Male	WA	Remote	10 plus	2020–21
008	30–39	Sarcoidosis	Male	VIC	Major cities	5–7	2015–19
009	under 20	PID	Female	QLD	Major cities	less than 0.5	2020–21
010	60–69	PID	Male	VIC	Major cities	10 plus	2010–14
011	60–69	PID	Female	SA	Major cities	10 plus	2005–09
012	50–59	Sarcoidosis	Female	ACT	Major cities	8–10	2005–09
013	20–29	PID	Female	VIC	Inner regional	8–10	2010–14

Phillips et al. Time to diagnosis for a rare disease: managing medical uncertainty. A qualitative study. doi [10.1186/s13023-024-03319-2](https://doi.org/10.1186/s13023-024-03319-2)

02 Why Current AI Fails

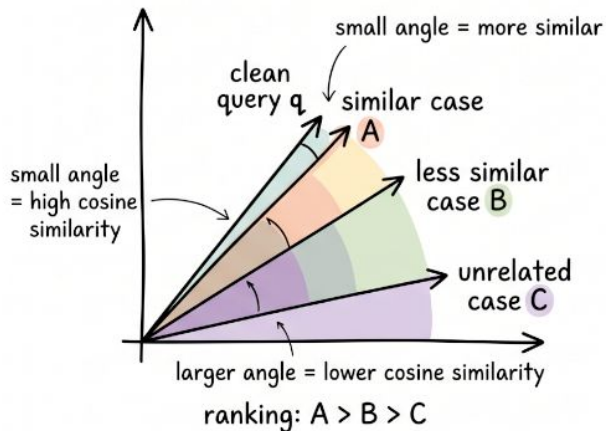
- (Centralized Big Data): Fails due to privacy risks and legal friction (DUAs). Documents in any form are not allowed to leave hospital.
- (Federated Learning): Fails due to "Catastrophic Forgetting." Generalizes data, erasing the exact "needle in the haystack" reference patients.

Inference-time learning on edge nodes is the only practical solution. Centralization moves sensitive data; model training compresses it. We want to search the evidence where it already lives.

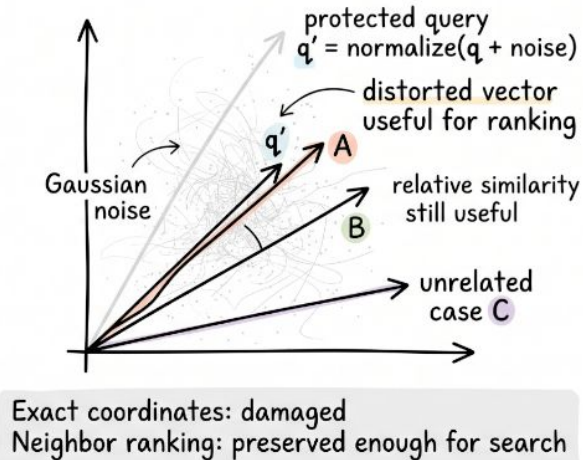
03 The FedRAG Paradigm Shift (1/2)

COSINE SIMILARITY RANKING

1. CLEAN VECTORS



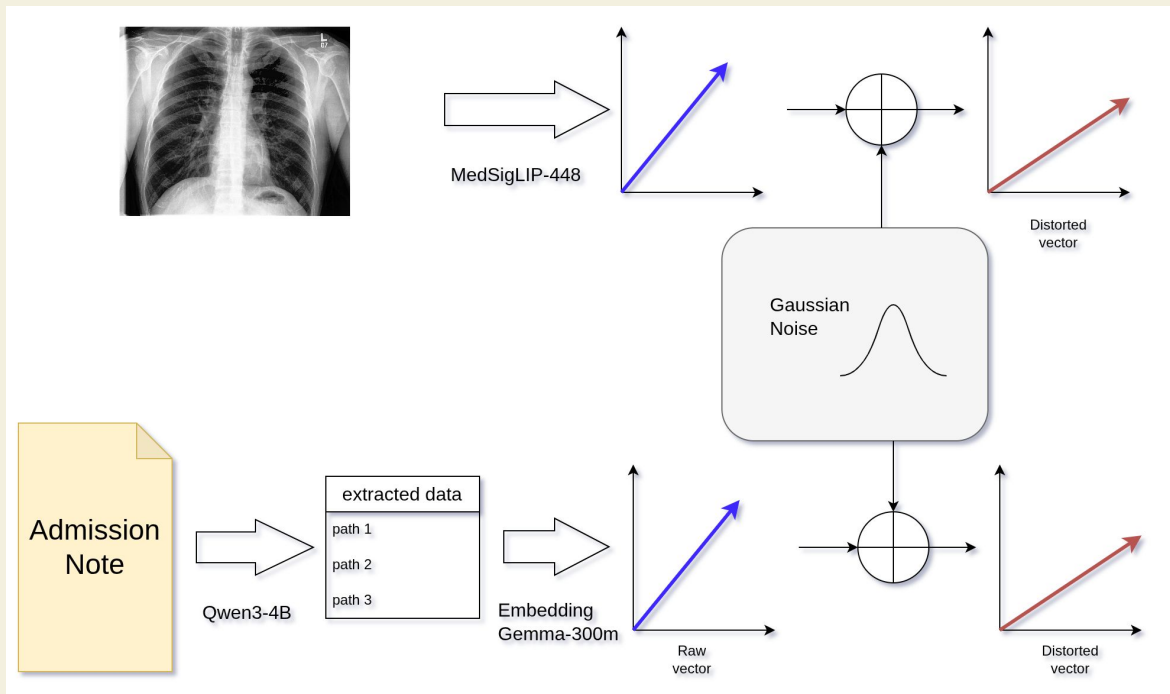
2. NOISY DISTORTION



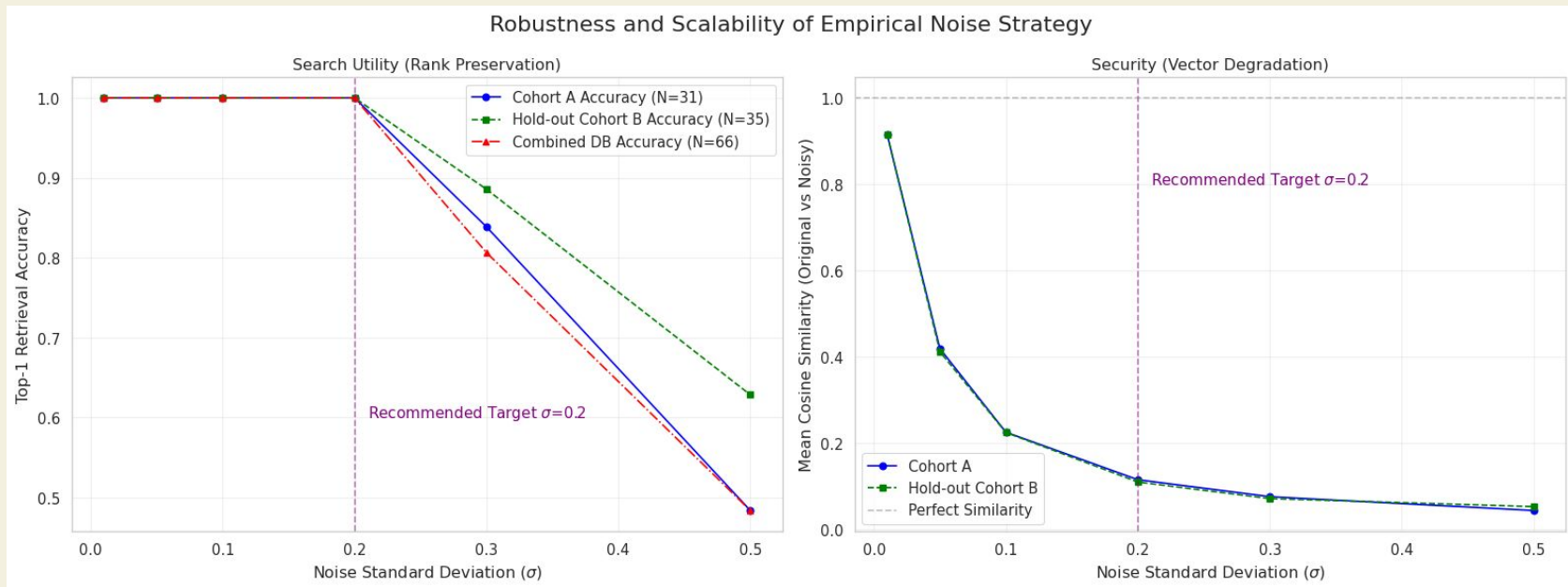
The vector gets ugly before the ranking gets useless.

04 The FedRAG Paradigm Shift (2/2)

Text and images become protected semantic signals staying what they are locally — raw clinical data never needs to become the query.

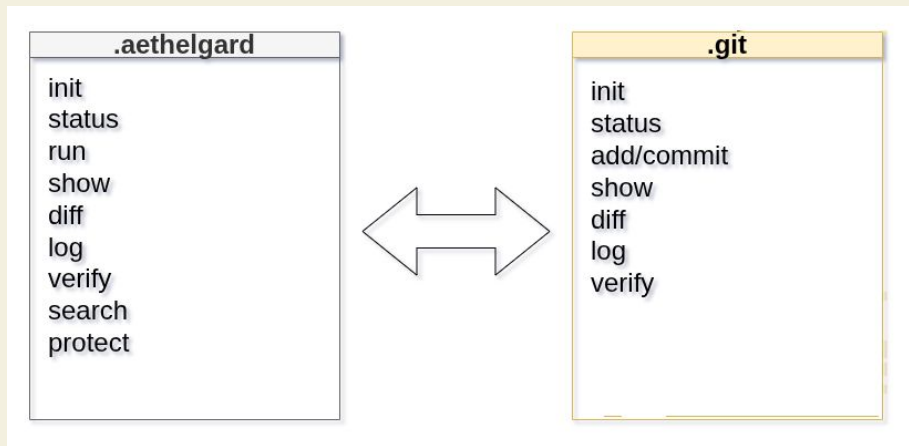
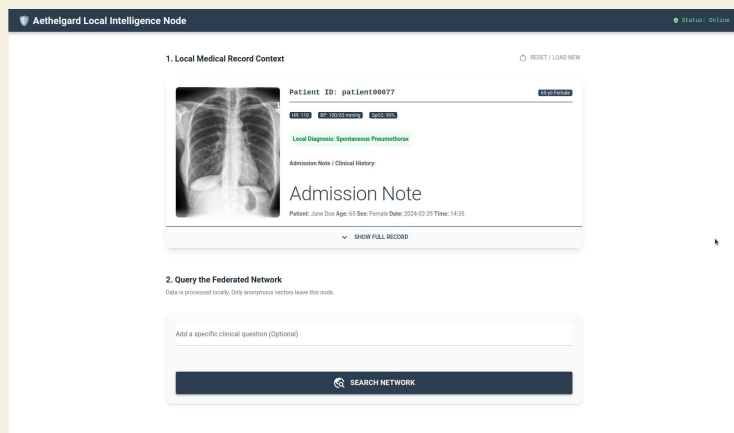


05 Semantic Firewalls and Generative Privacy



The useful trick is: representation degrades **faster** than retrieval quality, we can make vector unsuitable for reverse engineering..

06 Technical implementation



Git-like vault for local node, open-ended architecture allowing to implement any UI (think github for data)

Project Aethelgard.

Move the question, not the patient data.

Sharing knowledge to save lives



Aliaksei Kaliutau

aliaksei.kaliutau24@alumni.imperial.ac.uk

<https://github.com/akaliutau/aethelgard2>

<https://akaliutau.github.io/>